

WASP-84b

R-1122

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-84_180203 · created 2026-07-30T21:59:40+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458152.7482 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.111 +/- 0.014
Transit depth (Rp/Rs)^2	1.23 +/- 0.31 [%]
Orbital Inclination (inc)	88.88 +/- 0.52
Airmass coefficient 1 (a1)	0.975 +/- 0.012
Airmass coefficient 2 (a2)	0.019 +/- 0.0072
Scatter in the residuals of the lightcurve fit is	1.23 %
Best Comparison Star	#2 - [266, 134]
Optimal Aperture	3.41
Optimal Annulus	4.05
Transit Duration (day)	0.125 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.111 ± 0.014 vs 0.1295 (deviation 1.16σ)
Duration fit vs published	0.125 d vs 0.1146 d (deviation 0.76σ)
Mid-time O–C	-5.8 min
Depth SNR	7.0
Residual scatter	1.23 %

Light curve

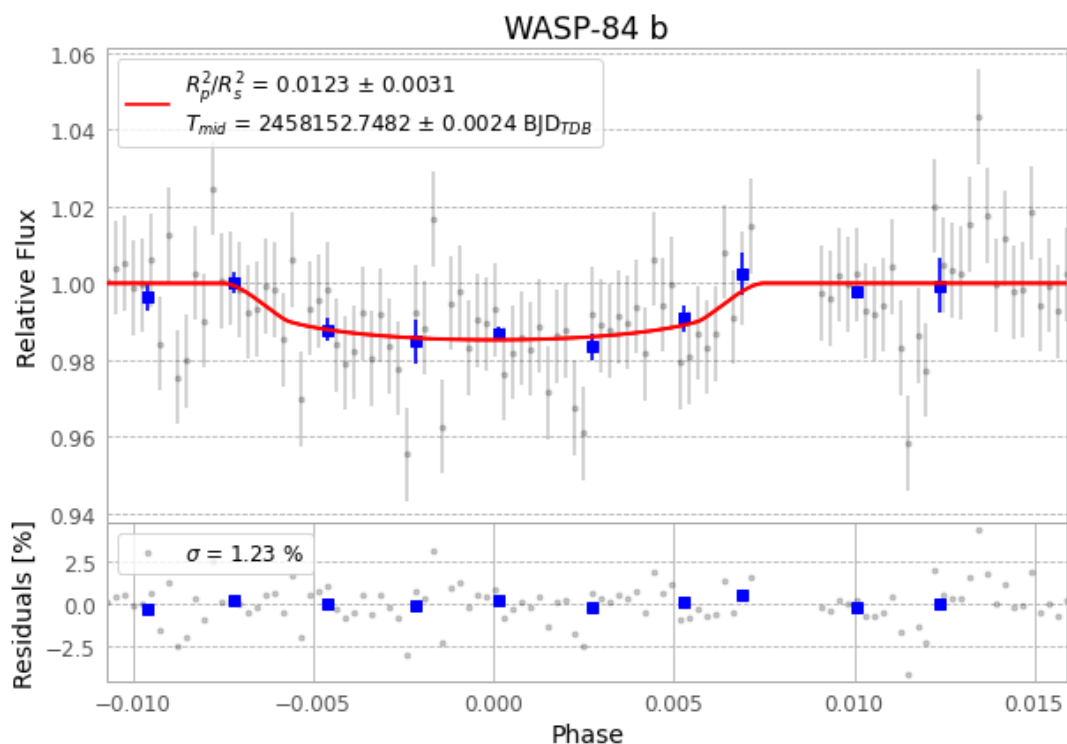


Figure 1 — FinalLightCurve_WASP-84 b_2018-02-02.png (SHA-256 25cde104cc6f474d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [317, 200], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 110f42b1b7a2644d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

110 FITS frames (72 MB), WASP-84180203033612.FITS → WASP-84180203090312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-84-180203.log (1f70cc1e7b97...), FinalLightCurve_WASP-84 b_2018-02-02.png (25cde104cc6f...), FinalLightCurve_WASP-84 b_2018-02-02.csv (6832ab0f7dc0...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 21:59Z.